

RadiSimCLIP: A Radiology Vision-Language Model Pretrained on Simulated Radiologist Learning Dataset for Zero-Shot Medical Image Understanding

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Abstract. Medical vision-language models (MVLMs) have shown significant promise in medical diagnosis. However, most existing approaches are constrained by limited modality coverage and often rely on datasets lacking real-world radiological diversity and semantic precision. Moreover, current models exhibit insufficient evaluation on fine-grained tasks. In this paper, we introduce **RadiSim**, a large-scale **Radiology** dataset comprising 10.6 million image-text pairs from CT, MRI, and DR modalities, designed to **Simulate** the learning resources during radiologist training. Building on RadiSim, we propose **RadiSimCLIP**, a radiology-specific MVLM pretrained using a CLIP-style contrastive framework. The model progressively acquires multilevel capabilities—ranging from anatomical recognition and pixel-level interpretability to vision-language alignment—mirroring how radiologists gain expertise. To systematically evaluate generalizability, we further propose a three-part evaluation framework: (1) zero-shot classification, which includes a novel extension to 3D volumetric data, evaluating anatomical, modality, and disease recognition; (2) zero-shot organ detection, which evaluates pixel-level semantic localization without parameter tuning; and (3) cross-modal retrieval, measuring alignment between visual features and clinical text. RadiSimCLIP achieves state-of-the-art results on 15 of 16 downstream benchmarks. This work underscores the feasibility and potential of large-scale radiology-specific MVLMs for robust, zero-shot 3D medical understanding, representing a practical step toward clinically applicable foundation models. The GitHub repository is <https://github.com/so-ux/RadiSimCLIP>.

Keywords: vision-language pre-training model · zero-shot classification · zero-shot detection · cross-modal retrieval

1 Introduction

Radiological imaging is indispensable in modern medicine; however, the increasing volume and complexity of scans have placed significant strain on radiologists, resulting in heightened workloads, interpretation delays, and inconsistencies. This situation underscores the pressing demand for robust automated analysis systems.

In recent years, medical vision-language models (MVLMs) have emerged as promising tools to address these challenges by integrating complementary features that enhance the efficiency and precision of medical services [2]. These models have demonstrated remarkable capabilities in various medical imaging tasks, including detecting chest pathologies from X-rays [19,20,23,25,30], diagnosing gastrointestinal diseases via endoscopic videos [22], assessing cardiac function using cardiac ultrasound videos [15], and analyzing histopathology images [7,8,13,14].

Despite these advances, current models primarily focus on specific diseases and modalities and still face significant limitations when applied to other modalities of radiological imaging. Although some existing methods can be adapted to general medical data, they still come with certain drawbacks. For example, PMC-CLIP [12] and BiomedCLIP [28,29] are based primarily on PubMed Central data (PMC), which do not adequately represent the complexities of medical imaging in the real world [10], limiting their sensitivity to specialized radiological terminology and visual characteristics. Additionally, while these models have been validated for image-text retrieval and 2D image classification, they are rarely extended to 3D volumetric applications. Other approaches [3,6,11,27] have expanded their datasets, but often lack validation for tasks requiring fine-grained recognition and interpretability, such as zero-shot detection. This gap hinders their effectiveness in high-precision medical tasks, where the ability to locate lesions or organs without additional fine-tuning is crucial. However, few studies have explored the capabilities of CLIP-based models for zero-shot detection in a medical context.

To address these challenges, we first introduce RadiSim, a large-scale, multi-modal radiology dataset comprising 10.6 million meticulously curated image-text pairs from CT, MRI, and DR modalities. The name "RadiSim" combines **R**adiology and **S**imulating training resources—designed to emulate the diverse materials encountered during radiologist education, such as anatomical references and pathological case descriptions. Through large-scale pretraining on RadiSim using contrastive learning (e.g., CLIP), we propose RadiSimCLIP, which progressively acquires hierarchical competencies—from basic anatomical recognition to diagnostic reasoning—mimicking the structured knowledge acquisition process in clinical training. We establish a multi-granular evaluation protocol: 1) Zero-shot classification (2D/3D) for organ/modality/disease recognition, 2) Pixel-level organ detection without fine-tuning, and 3) Cross-modal retrieval for semantic alignment verification. RadiSimCLIP achieves state-of-the-art performance on 15 out of 16 radiological downstream tasks, demonstrating its strong generalization ability and potential as a universal radiology assistant.

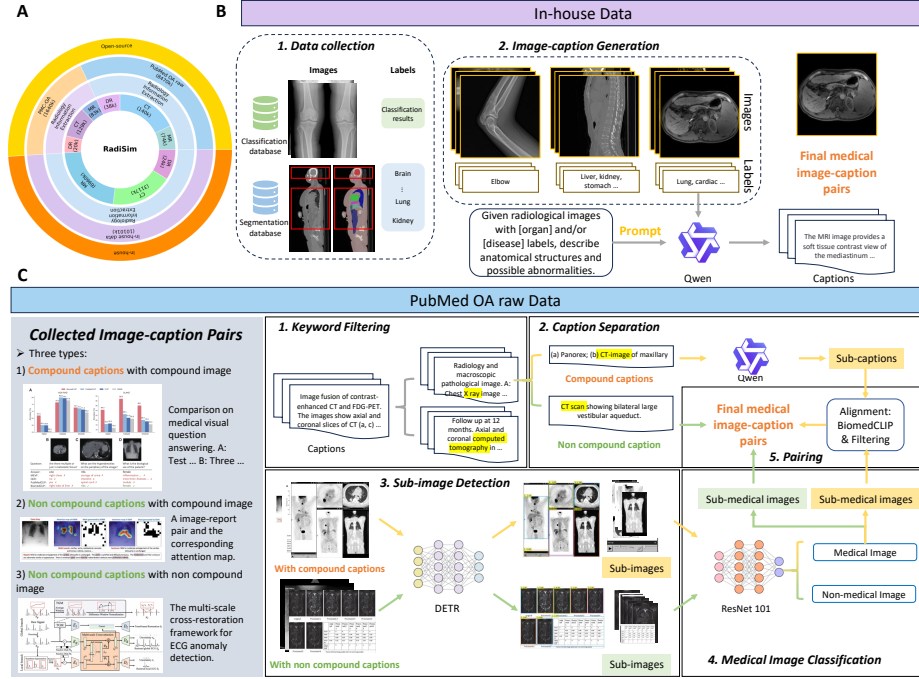


Fig. 1: Construction of the RadiSim dataset. (A) Overview of RadiSim, encompassing both in-house and publicly available data sources; (B) Workflow for constructing the in-house dataset; (C) Data curation pipeline for the PubMed OA raw dataset.

2 Construction of The RadiSim Dataset

To address the lack of representativeness and diversity in existing datasets, we constructed a comprehensive image-caption pair dataset named RadiSim. This dataset incorporates 10,585k in-house samples and 484k curated samples from diverse open-source datasets sourced from two public resources: PubMed Central (PubMed OA raw) [16] and PMC-OA[12] (Figure 1A).

The in-house portion (Figure 1B) is built via a two-stage pipeline: data curation and caption generation. We first collect a diverse set of images and associated disease or organ labels from internal classification and segmentation tasks across partner hospitals. While segmentation samples are limited, they help identify regions of interest, enhancing anatomical precision. Next, a domain-adapted large language model (Qwen2 [21]) is prompted with both image content and structured labels to produce detailed, anatomically grounded captions. This results in rich, aligned image-text pairs with high semantic fidelity. For PubMed OA raw, our data curation process (Figure 1C), inspired by PMC-CLIP, categorizes content into three types: compound captions with compound images, non-compound captions with compound images, and non-compound captions

with non-compound images. These types undergo several steps: 1) keyword filtering to select relevant radiology image-caption pairs; 2) caption separation using Qwen2 to decompose compound captions into sub-captions; 3) sub-image detection and cropping for compound images, 4) ResNet101 is employed to filter out non-medical images; and 5) consolidation of all qualifying pairs into the dataset, resulting in 202k radiological image-text pairs. For the PMC-OA dataset, we filtered out non-radiological pairs, retaining 232k relevant image-text pairs.

3 Experiments

3.1 RadiSimCLIP Training

We fine-tuned the pre-trained BiomedCLIP model on our RadiSim dataset to enhance its adaptation to radiological imaging (Figure 2). Training followed a two-stage strategy to leverage data heterogeneity: we first warmed up the model on the external public subset to encourage general cross-modal alignment across diverse imaging styles, then fine-tuned it on the in-house dataset containing more clinically grounded semantics. The in-house data includes classification-derived pairs with anatomical and diagnostic labels, and segmentation-derived samples emphasizing organ-level spatial cues.

We adopted a selective freezing strategy, keeping most of the text encoder frozen except for LayerNorm parameters, while fully training the visual encoder and projection layers. The model was trained using AdamW with a batch size of 4096 for 30 epochs. Training was conducted on 8 NVIDIA L40 GPUs and took approximately one week to complete.

3.2 Downstream Datasets

To comprehensively evaluate the performance of RadiSimCLIP in anatomical understanding, fine-grained recognition, and general medical reasoning, we designed and conducted four downstream experiments across diverse datasets, as detailed in Table 1.

3.3 Downstream tasks

2D Zero-shot Classification We first assess RadiSimCLIP’s zero-shot capabilities on 2D medical images. For CT imaging, we conduct 2D slice-based eight-class classification across three anatomical planes (axial, sagittal, and coronal), while for DR imaging, we evaluate on PneumoniaMNIST [26] and RSNA Pneumonia [17] datasets. The model performs zero-shot classification by computing cosine similarity scores between image embeddings and pre-defined class-descriptive text embeddings, eliminating the need for fine-tuning. We compare our approach with state-of-the-art models including CLIP [18], MedCLIP [23], PubMedCLIP [4], PMC-CLIP, and BiomedCLIP.

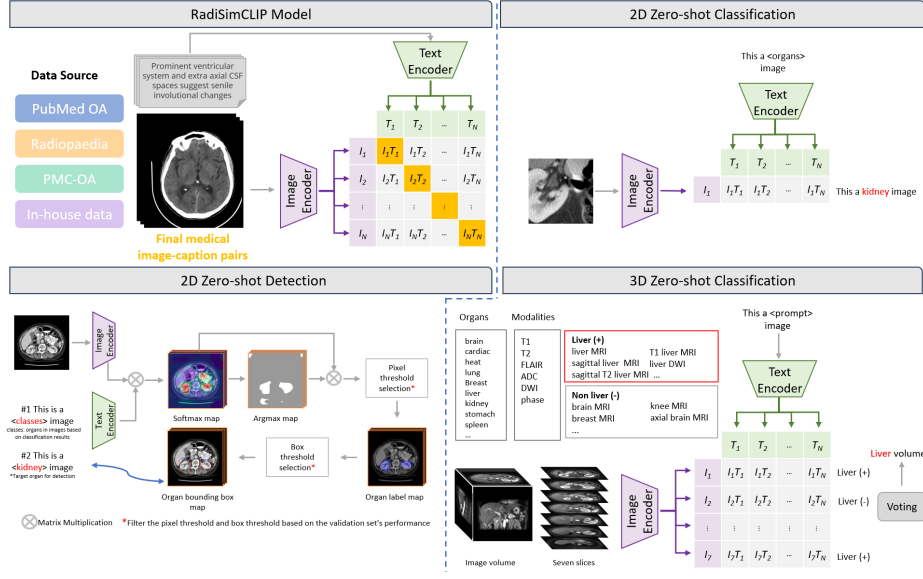


Fig. 2: Overview of RadiSimCLIP training and three typical downstream tasks.

3D Zero-shot Classification Our 3D volume zero-shot classification tasks are performed on the in-house dataset and open-source datasets (i.e., knee MRI ACL and Meniscus [1]), using a three-stage reasoning framework. Specifically, in the image embedding construction phase, we evaluated multiple 2D slice sampling strategies (e.g., median slice, maximum intensity slice, and uniform sampling) to balance computational efficiency and spatial coverage. Ultimately, we selected uniform sampling across seven slices along the z-axis to extract anatomical features. In the semantic constraint construction stage, we adhered to the T/CHIA 23-2021 standard classification codes for medical imaging examination sites to establish a constrained text generation paradigm. This involved creating a foundational template for organ-modal combinations (e.g., “The modality image shows [organ]”) and employing a greedy algorithm to maximize the cosine similarity difference between positive and negative texts within the pool. During the decision-making fusion stage, we employed majority voting to aggregate the classification results of the volumetric data across N slices (in this work, $N = 7$).

2D Zero-shot Detection To evaluate the accuracy of text-guided visual models in identifying fine-grained information, we constructed a zero-shot detection task framework (Figure 2). Inspired by attention-based interpretation method [5], this framework decodes spatial attention patterns from the CLIP image encoder to localize anatomical structures. Specifically, we leverage text prompts describing organs to generate multi-label softmax attention maps through 12 attention heads in the final layer of the image encoder. These multi-head attention maps are aggregated to highlight regions semantically aligned with the input text. Using

Table 1: Overview of downstream tasks and evaluation datasets.

Task	Dataset	Metric	Description	Scale
2D Zero-shot Classification	CT-Axial	ACC	Eight-class organ classification (MediMeTA[24]), with bilateral organs merged	1,645 images
	CT-Sagittal	ACC		1,645 images
	CT-Coronal	ACC		1,645 images
	Pneumonia-MNIST[26]	AUC	Pneumonia <i>vs.</i> normal in chest X-rays	5,856 images
	RSNA Pneumonia[17]	AUC	Pneumonia <i>vs.</i> normal in chest X-rays	23,286 images
3D Zero-shot Classification	MRI organ Brain	ACC	Brain <i>vs.</i> other organs	63,910 volumes
	MRI organ Breast	ACC	Breast <i>vs.</i> other organs	63,910 volumes
	MR modality (TOF-MRA)	ACC	MR angiography (MRA) <i>vs.</i> conventional MRI	5,688 volumes
	MR modality (T1C)	ACC	T1 contrast <i>vs.</i> non-contrast classification	5,076 volumes
	Knee MRI disease (ACL)[1]	AUC	ACL tear <i>vs.</i> non-ACL tear using sagittal view	1,250 patients
	Knee MRI disease (Meniscus)[1]	AUC	Meniscal tear <i>vs.</i> nonmeniscal tear using sagittal view	1,250 patients
2D Zero-shot Detection	In-house Dataset	F1	Nine organ detection in MR volumes	1,117 detection boxes
	AMOS-CT[9]	F1	Seven organ detection in CT volumes	8,289 detection boxes
Cross-modal Retrieval	PubMed OA-MR	Top-1%	Text-to-image retrieval and image-to-text retrieval	26,589 images
	PubMed OA-CT	ACC,		18,008 images
	PubMed OA-DR	Avg. Rank		4,503 images

a predefined organ vocabulary, we then visualize the corresponding activation maps and apply a thresholding strategy to suppress low-confidence responses. Finally, bounding boxes are derived from the thresholded activation maps to localize target organs in the image.

During the experimental phase, we employed this framework for anatomical localization of 13 representative organs and a background class (i.e., brain, stomach, lung, heart, breast, liver, spleen, kidney, pancreas, prostate, bladder, uterus, knee, and others). Text embeddings for these 14 categories were generated via a pre-trained text encoder, then projected into a shared embedding space with image embeddings. Relative confidence scores were computed by normalizing cosine similarity, followed by softmax activation. To obtain accurate organ localization, we applied the threshold corresponding to the best F1 score calculated on the validation data set to generate a binary activation map on the test set. Subsequently, connected component analysis was performed on the thresholded maps to extract bounding boxes for specified organs.

Cross-Modal Retrieval We utilized untrained data from PubMed-OA to assess the degree of the model correlating medical images with their corresponding textual descriptions and vice versa.

4 Results

4.1 2D Zero-shot Classification

Table 2 presents a comprehensive comparison of 2D zero-shot classification performance across different models on medical imaging tasks. These results collectively demonstrate that RadiSimCLIP excels in both modality-specific recognition and lesion detection, outperforming other baseline models by significant margins. Specifically, RadiSimCLIP achieves remarkable performance on CT imaging across all three anatomical planes, attaining accuracy scores of 0.765, 0.683, and 0.711 for axial, sagittal, and coronal views, respectively. In DR tasks, RadiSimCLIP exhibits state-of-the-art performance on PneumoniaMNIST with an accuracy of 0.728, while BiomedCLIP slightly outperforms on the RSNA dataset with an accuracy of 0.765. In contrast, the general-purpose CLIP model shows limited effectiveness, achieving only 0.106 accuracy on CT tasks and random-chance performance (0.500 AUC) on DR tasks.

Table 2: Performances of zero-shot classification on different modalities and tasks.

Model	CT			DR		MRI organ		MR modality		MRI disease	
	Axial	Sag.	Cor.	MNIST	RSNA	Brain	Breast	MRA	T1C	ACL	Meniscus
CLIP	0.106	0.106	0.106	0.500	0.500	0.787	0.716	0.155	0.908	0.620	0.582
MedCLIP	0.261	0.292	0.314	0.498	0.711	0.763	0.888	0.270	0.852	0.534	0.551
PubMedCLIP	0.419	0.343	0.398	0.553	0.646	0.999	0.944	0.864	0.949	0.445	0.509
PMC-CLIP	0.300	0.272	0.277	0.556	0.579	0.909	0.803	0.647	0.937	0.618	0.427
BiomedCLIP	0.582	0.617	0.644	0.689	0.765	1.000	0.998	1.000	0.998	0.451	0.430
RadiSimCLIP	0.765	0.683	0.711	0.728	0.761	1.000	0.999	1.000	1.000	0.707	0.643

4.2 3D Zero-shot Classification

We validated the classification capability of RadiSimCLIP for organs, modalities, and diseases in MR data, with the results presented in Table 2. In the organ binary classification task, RadiSimCLIP achieved outstanding performance, attaining an accuracy close to 1.000. Regarding modality classification, RadiSimCLIP also exhibited high accuracy. In the disease multi-classification task, RadiSimCLIP achieved the highest score in the AUC metric, with scores of 0.707 for ACL tears and 0.643 for meniscus tears, significantly outperforming other models. These results suggest that RadiSimCLIP possesses exceptional capabilities in identifying disease-related patterns within zero-shot MRI data and demonstrates a superior understanding of image details compared to alternative models.

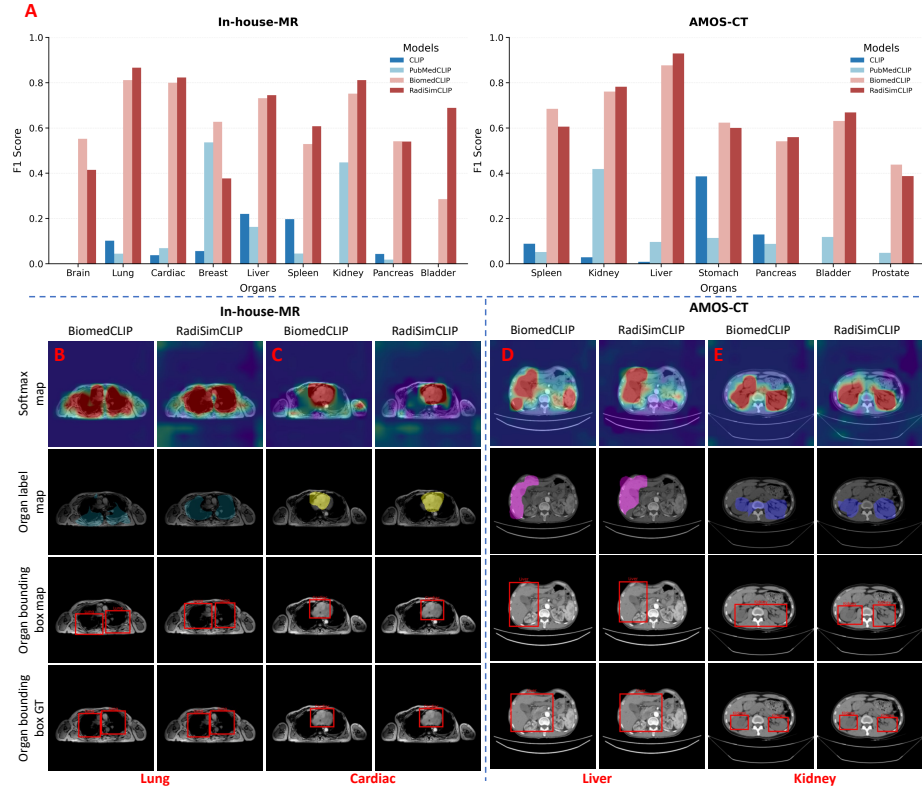


Fig. 3: Results of zero-shot detection across various organs in different datasets.

4.3 2D Zero-shot Detection

We applied our zero-shot detection framework to CLIP, PubMedCLIP, BiomedCLIP, and RadiSimCLIP (Figure 3). From Figure 3A, RadiSimCLIP achieved the highest scores in seven out of nine organ categories on internal datasets, and showed superior performance on most organ categories in the AMOS-CT dataset. Due to the limited softmax map content in CLIP and PubMedCLIP, we focus our visual comparison on BiomedCLIP and our method. Our results demonstrate three key advantages: (1) more precise organ localization, with softmax maps concentrated on target organs, (2) improved alignment with organ morphology, closely matching ground-truth boundaries, and (3) accurate organ number detection through threshold-based feature screening, beneficial particularly for paired organs such as the lungs and kidneys.

4.4 Cross-Modal Retrieval

As shown in Table 3, our proposed RadiSimCLIP demonstrates superior performance in cross-modal retrieval tasks across all medical imaging modalities

Table 3: Performance of cross-modal retrieval.

Dataset	BiomedCLIP		RadiSimCLIP	
	Top1% Acc $\uparrow$	Avg. Rank $\downarrow$	Top1% Acc $\uparrow$	Avg. Rank $\downarrow$
PubMed OA-MR	0.428	2016	0.763	523
PubMed OA-CT	0.698	457	0.781	310
PubMed OA-DR	0.655	111	0.717	97

compared to BiomedCLIP. Most notably, on the PubMed OA-MR dataset, our method achieves a substantial improvement with a Top1% accuracy of 0.763 (*vs.* 0.428) and reduces the average rank from 2016 to 523. Similar improvements are also observed in CT (0.781 *vs.* 0.698) and DR (0.717 *vs.* 0.655) modalities, with consistently lower average ranks.

5 Conclusion

We have constructed RadiSim, a radiology-specific dataset with 10.6M multi-source image-text pairs. Using RadiSim, RadiSimCLIP achieves SOTA accuracy on 15/16 tasks via hierarchical validation: 1) anatomical recognition via zero-shot classification, 2) pixel-level understanding through zero-shot detection, and 3) semantic alignment via cross-modal retrieval. This demonstrates general-purpose AI’s potential for specialized medical imaging.

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